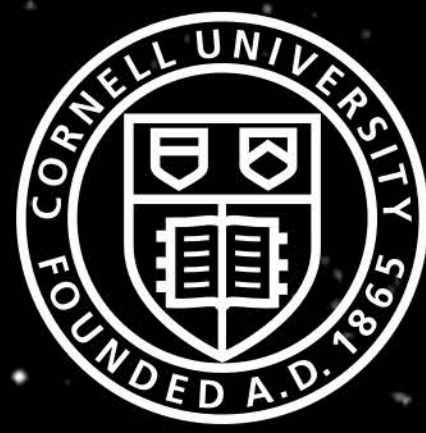


Virtual Reality as a Teaching Tool for Moon Phases and Beyond

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References

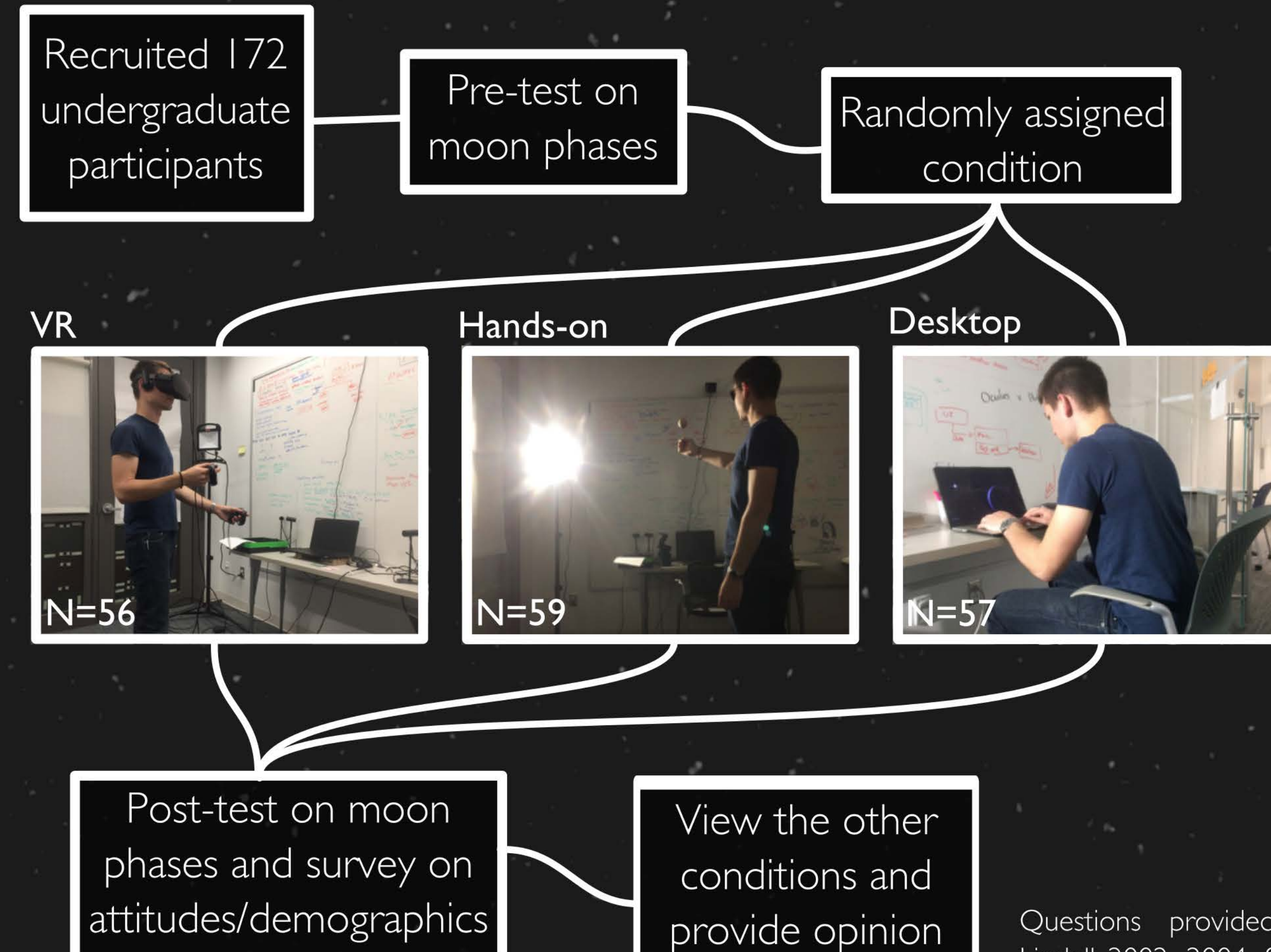


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Introduction

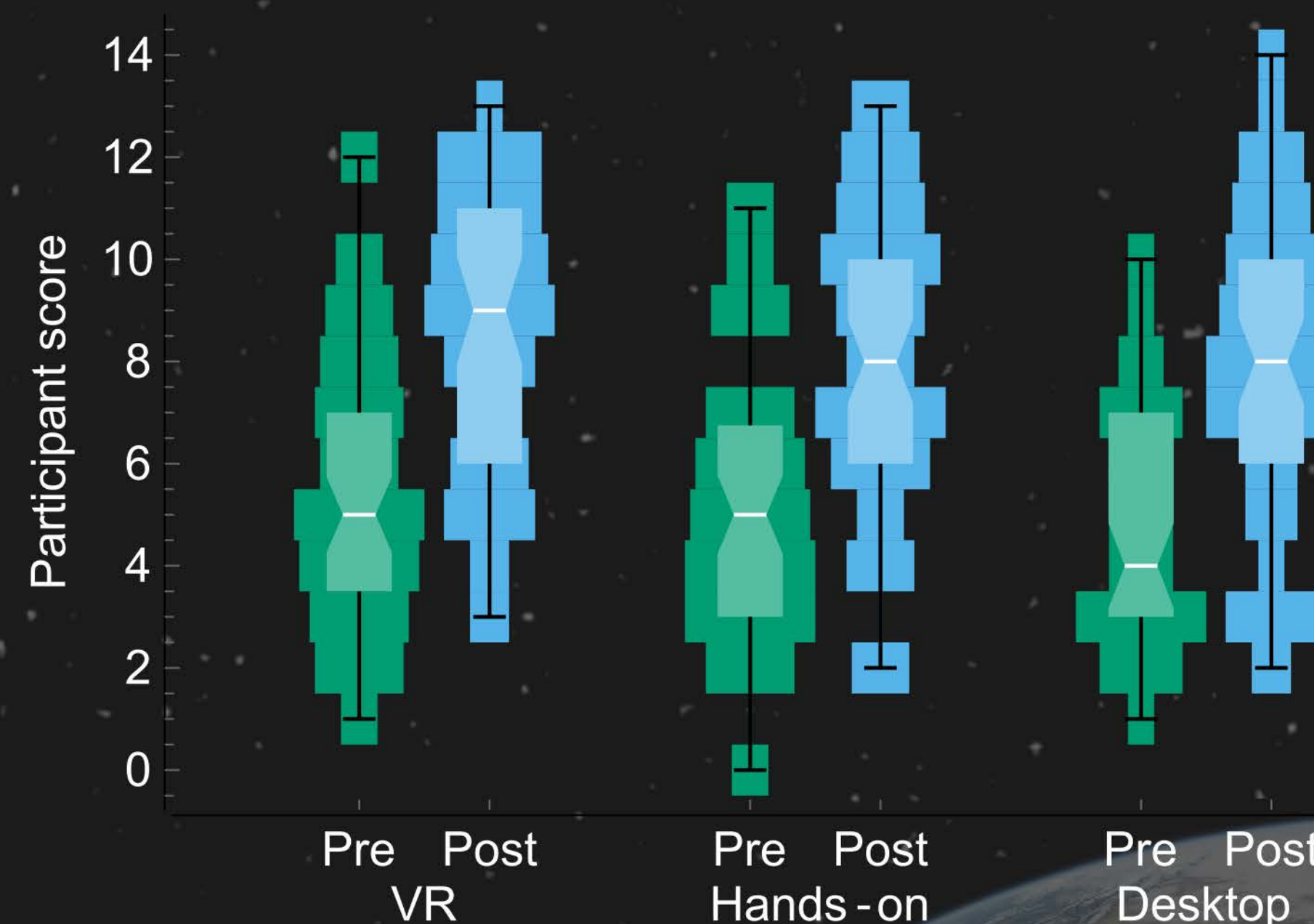
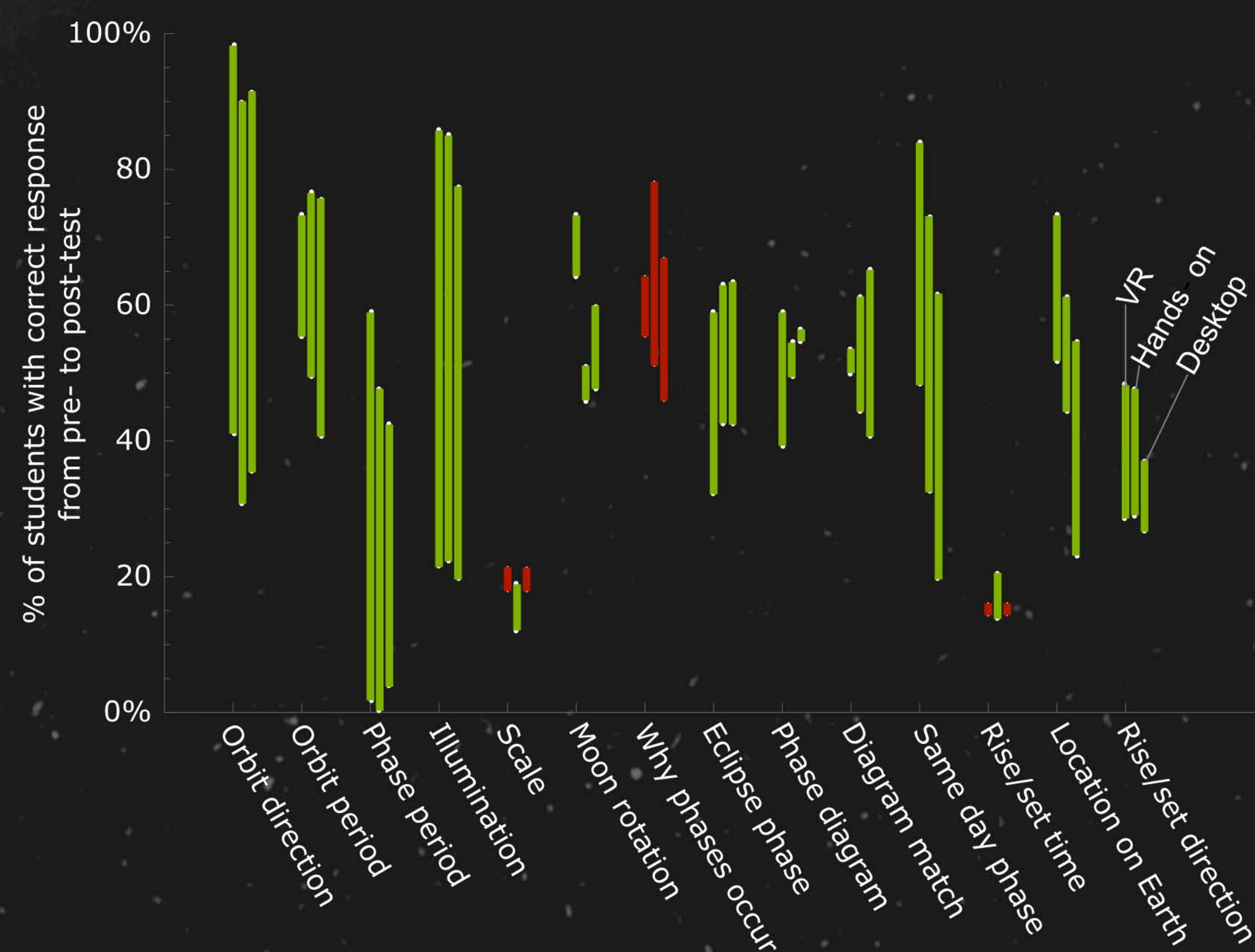
In an educational context virtual reality (VR) provides a classroom with the embodied learning of a hands-on activity and the digital versatility and consistency of a computer simulation. This makes immersive VR a potentially effective teaching tool, but not much research has been done to test it. **We used the intro astronomy subject of moon phases as the topic to measure the learning gains between a self-guided lesson in VR and traditional lesson activities.** We found a strong participant preference for VR relative to the traditional methods. However, we observed no difference across conditions in average levels of performance on a pre/post knowledge test.

Method



Result 2: Pre/post test scores by question

When looking at the performance from pre to post test broken down by topic we see that each condition was similarly capable of conveying the concepts. This chart shows performance gains (green) and losses (red) of the participant pool broken up by condition on each question topic.



Result 1: Pre/post test scores

Across the three conditions there was a marked improvement in participant performance from the pre-test to the post-test, an average jump of 22%. Between the conditions there was no significant difference in the post-test score meaning the VR condition taught students just as well as the traditional methods.

Result 3: Condition Preference

After the post-test, participants saw the other two conditions they hadn't seen and were asked their preference. There is a striking preference for VR.

Which of the three ways to simulate moon phases is your favorite?

78.2%

VR

10.0%

Hands-on

11.2%

Desktop

Conclusions and Future Work

By examining the learning gains of participants after lessons utilizing virtual reality and traditional methods we find no significant differences except for personal preference, which strongly favored VR. **This work shows that even at VR's early stage of development and adoption it is capable of teaching the content as well as current tried-and-true methods and with higher student interest.** Further analysis needs to be done on any demographic correlations to performance such as video game experience and previous VR use.

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